

Optimal Hallway-Based Fire Extinguisher Placement in the CCIS Building Using Integer Programming

**Ladera, Ivan Justin L.
Flores, Gian Rusell
Natividad, John Lloyd James**

College of Computing and Information Sciences
Bachelor of Science in Computer Science

Problem Description

Introduction

Fire safety is an essential concern in buildings where many people are present, such as schools. In emergency situations, fire extinguishers play a crucial role in controlling small fires and preventing further damage. However, determining the appropriate number and placement of fire extinguishers within a building can be challenging. Placing too many units may lead to unnecessary costs, while placing too few may result in poor accessibility during emergencies.

To address this problem, proper planning is required to ensure that fire extinguishers are both sufficient and easily accessible from different areas of the building. This study aims to determine the optimal placement of fire extinguishers in the CCIS building using Integer Programming. The objective is to minimize the number of fire extinguishers while ensuring that all areas of the building remain accessible through strategic placement.

Context of the Problem

The CCIS building is a two-floor structure composed of hallways, lobbies, and staircases that serve as common access points for occupants. In practice, fire extinguishers are typically installed along hallways because these areas provide central access to nearby rooms, restrooms, lobbies, and staircases.

For modeling purposes, this study focuses on hallway sections as the primary coverage areas. Hallways are considered representative of overall accessibility since they serve as the main pathways used by occupants during normal conditions and emergencies. This approach simplifies the model while still ensuring that all important areas of the building remain accessible.

However, determining the appropriate placement of fire extinguishers remains a

challenge. Installing too many units can lead to unnecessary costs, while installing too few may result in insufficient coverage. Therefore, there is a need to identify the most efficient placement strategy that ensures full accessibility while minimizing the number of units required.

To address this, the study applies Integer Programming using a Set Covering Model to determine the optimal locations for fire extinguisher placement within the building.

Research Questions (Experiment Questions)

This project aims to answer the following questions:

1. What is the minimum number of fire extinguishers required to ensure accessibility across all hallway areas of the CCIS building?
2. Where should the fire extinguishers be placed to achieve optimal coverage?
3. How can the placement be optimized while minimizing the number of units installed?

